Qualcomm Technologies, Inc.

# SW6100 Linux Android Wear (LAW1.0) Current Consumption Data

## Application Note

80-83950-7A Rev. AD

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hyungchul.won@lge.com

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# Revision history

| Revision | Date | Description |
|---|---|---|
| AD | October 2025 | Addded Section 4, *Chipset current consumption* |
| AC | September 2025 | Updated Chapter 3, *Test definitions* |
| AB | August 2025 | Updated power gird in Section 2.1, Power tree |
| AA | July 2025 | Initial release |

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hyungchul.won@lge.com

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# 1 Introduction

This document provides current consumption data for SW6100 Linux Android Wear 1.0. The current consumption is highly dependent on software optimization.

## 1.1 Conventions

Function declarations, function names, type declarations, attributes, and code samples appear in a different font, for example, `cp armcc armcpp`.

Code variables appear in angle brackets, for example, `<number>`.

Commands to be entered appear in a different font, for example, **`copy a:*.* b:`**.

Button and key names appear in bold font, for example, click **Save** or press **Enter**.

Mozilla Firefox is the recommended browser to view attachments.

## 1.2 Expectations

### Device variations

The current consumption measurements recorded in this document are not expected to match the current consumption measurements of a customer design. They are also not expected to exactly match the current consumption measurements of another wearable test platform (WTP) of identical part numbers loaded with identical software.

Variations in measurement, when compared to the customer device, are caused by the WTP using different components than the customer design.For example, memory ICs, display ICs, peripheral ICs. Variations in measurement when compared to an identical WTP loaded with identical software are caused by normal silicon process variations in both the Qualcomm Technologies, Inc. (QTI) and non-QTI components and power supply tolerances. Any differences in measurement technique, equipment, or temperature also cause variations. Other factors, for example, floating CMOS inputs or taking measurements at maximum Tx power, can affect both the reliability and repeatability of current consumption measurements.

Targets and measurements contained in this document are typical values measured on a single WTP in a lab bench environment (room temperature). They are provided as a relative reference point, not as an absolute goal to attain.

### Test case selection

Test cases selected for measurement in this document are intended to provide a wide range of coverage. Although the specific conditions a customer needs may not appear in this list, there is generally a test case or combination of test cases that is close enough to the customer requirement to be used for a baseline comparison.

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nyu.mobile.wonders@gmail.com

## OEM platform power optimization

OEM platform power optimization starts by comparing the QTI platform against one or more baseline test conditions, as mentioned in this document.

1. The OEM and SW6100 platforms are compared and optimized in a known test case.
2. Then the differences between baseline and required test cases are measured on the OEM platform and analyzed with QTI for optimization.

Power optimization of a customer platform is an iterative process. The process involves:

1. Identifying power consumption differences between a customer platform and QTI platform in similar test conditions.
2. Determining the source of those differences; these could be test conditions, hardware configuration differences, software control differences, and so on.
3. Deciding whether the source is an error to be be corrected, or is intentional.
4. Correcting errors that are identified.
5. Repeating until all differences are corrected or determined to be intentional.

As customers update their hardware platform or integrate new software releases from QTI, power consumption may change, triggering additional power optimization. It is important to understand that during the software development process, current consumption for some test cases may increase as power improvement features may need to be delayed to a later release to meet stability requirements.

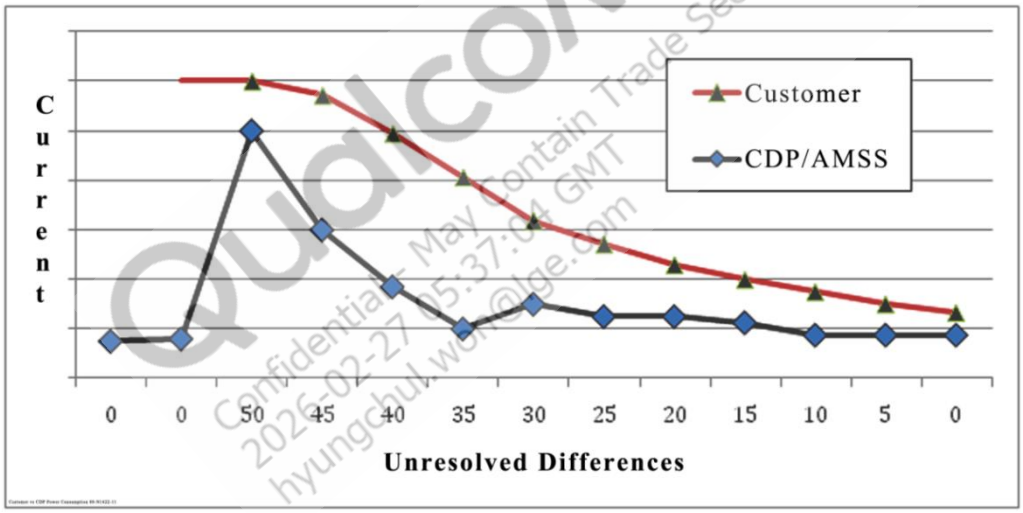


**Figure 1-1 Representation of customer vs. WTP power consumption**

When discussing power optimization with QTI, provide the following information:

- Specific test conditions
- Available history of power measurements for these test conditions
- Goals for power optimization
- Recent hardware or software changes that may affect power consumption
- Power consumption breakdown per voltage rail, if possible

## 1.3 Publication timeline for power measurement data

This document is updated monthly with the latest available current consumption goals and battery-level measurements. Following the commercial sample (CS) software release, key power rail breakdowns are included as they become available. Measurement of these power rail breakdowns depends upon customer requests for specific test cases and software releases.

**Table 1-1 Timeline for publishing current consumption data**

| Information provided | Guideline for availability |
|---|---|
| Target current consumption for commercial software release | Available at the time of hardware engineering samples |
| Battery-level measured values for Table 3-1 test cases | Available for CS software release and for major software releases thereafter; these measurements can also be found within the software release notes |
| Key power rail breakdowns for Table 3-1 est cases | Provided on request for CS and later software releases; submit requests at https://www.qualcomm.com/support |

## 1.4 GPIO configuration

To minimize current consumption, the user must configure any unused GPIOs in one of two ways:

- As outputs in their Logic Low state
- As inputs with their internal pull-downs enabled

1. Any GPIO pin configured as an input is driven by a peripheral device (Bluetooth, WLAN, NFC, and so on). The device must be programmed with an internal pull-down, whenever the corresponding peripheral signal is set to its High Impedance state.
2. The GPIO should then be reprogrammed to remove the pull-down when the peripheral signal is taken out of its High Impedance state.

   These steps are required to prevent unwanted oscillation or high current leakage on the device's pins. For additional information, see *Configuration of Input Pins During Device Sleep* (80-VN499-7).

## 1.5 Technical assistance

For assistance or clarification on information in this document, go to: https://www.qualcomm.com/support

For assistance through Salesforce case, select the problem area as listed in the following table:

**Table 1-2 Power consumption measurement for core power and multimedia**

| Problem area | Core power | Multimedia |
|---|---|---|
| **Problem area 1** | **BSP/HLOS** | **Multimedia** |
| **Problem area 2** | **Power (SW/HW)** | **Power** |
| **Problem area 3** | OEM appropriate selection | OEM appropriate selection |

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# 2 Test setup

### Devices used for testing

The following hardware and software configurations are used for the measurements:

- The SW6100 chipset and PMW6100, installed on a specially configured resistive current measurement (RCM) evaluation platform for power measurement purposes only.
  - MSM™ Chipset: SW6100 TT part
  - Memory: 4 GB LPDDR5X + 64GB EMMC
  - RF card: SDR125
  - PMIC: PMW6100
- Various SW6100 software releases as defined throughout this document, along with the current consumption data being presented.
- Display resolution configured at 480 x 480 smart panel.

Table 2-1 lists the chipset's device revisions used on the SW6100 power measurement platform for each set of tests. Throughout this document, the phrase *SW6100 chipset* means this combination of devices.

**Table 2-1 Chipset ICs tested**

| QTI IC MSM configuration |
|---|
| SW6100 |
| PMW6100 |
| SDR125 |
| QCC6100 |
| With SIM card |
| With sensor bar |

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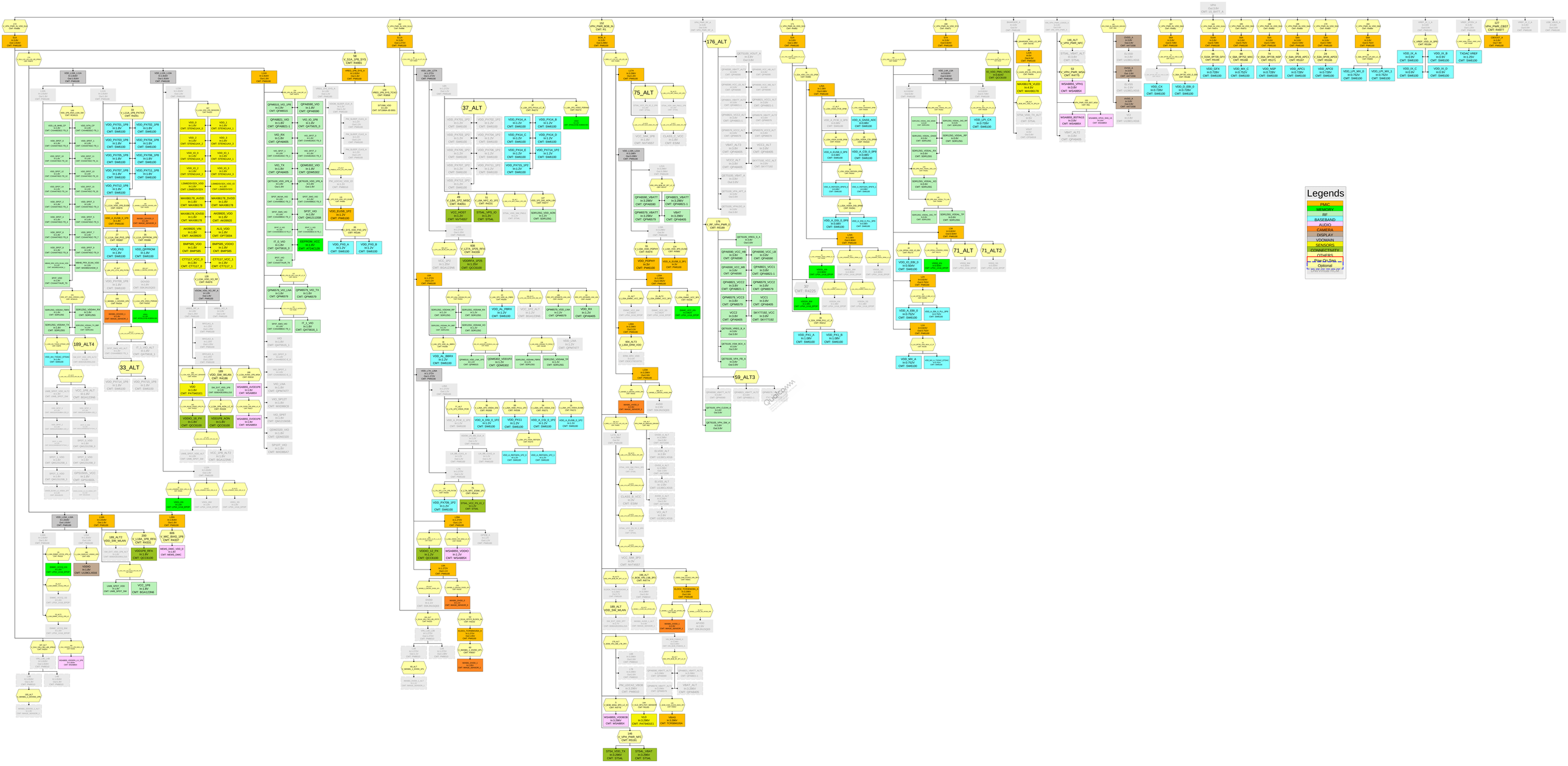
## 2.1 Power tree

The power tree for the SW6100 WTP-DVT shows the power rails for devices such as the QCC6100 and external circuitry. The external circuitry includes: eMMC, SD card, display, and sensors.

The current consumption numbers listed in the power tree are estimated worst-case currents. These numbers are preliminary and are intended for PCB routing reference only. In a real-world scenario, all the loads at a node do not run concurrently with peak consumption.




SW6100_WDP6100 Power Grid

# 3 Test definitions

Definitions of test conditions (air interface configuration, multimedia, display activity, lighting status, and so on) for the current consumption tests are listed in the following table.

**Table 3-1 Test definitions**

| Test case | Case ID | Test conditions |
|---|---|---|
| MSM Airplane mode sleep With display, Sensors, KWD , audio , tilt , touch, ALL OFF , DDR in SR | AIR1 | Settings:<br>1. Airplane mode: Turn OFF Bluetooth/Wi-Fi/UWB/GPS location and enable the airplane mode.<br>2. Display: AON disable, Tilt to bright (TTB): Disable, Tilt to wakeup (TTW): disable, display OFF<br>3. Sensor: Disable.<br>4. Touch panel: Disable.<br>5. Offload: Disable.<br>6. For required configurations, see Other settings.<br>Measure:<br>Disconnect USB and wait for device suspend then start power measurement. |
| Display ON (static display with smart panel) | LCD04A-C | Test application:<br>1. Install apk: `com.qualcomm.qti.weartech.watchface.test`.<br>2. Select the digital, black and white font watchface, which updates every minute on UI application list.<br>Settings:<br>1. Do the AIR1 settings.<br>2. AON and Offload: enable. For required configurations, see<br>3. Other settings<br>4. Display: Turn ON display AON.<br>5. Offload: Enable offload.<br>Measure:<br>Disconnect USB and wait for display time out and measure power for 30 sec ( Don't include minute update) |

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| Test case | Case ID | Test conditions |
|---|---|---|
| Static display with clock update once a minute | LCDCU1 | Test application:<br>1. Install apk: `com.qualcomm.qti.weartech.watchface.test.`<br>2. Select the digital, black and white font watchface, which updates every minute in UI application list.<br>Settings:<br>1. Do the AIR1 settings.<br>2. AON and Offload: enable. For required configurations, see Other settings.<br>3. Display: Turn ON display AON.<br>4. Offload: Enable offload.<br>Measure:<br>Disconnect USB and wait for display time out and measure power for 60 sec (Include minute updates) |
| Static Display with Clock Update once a second | LCDCU60 | Test application:<br>1. Install apk: `com.qualcomm.qti.weartech.watchface.test.`<br>2. Select the digital, black and white font watchface, which updates every second in UI application list.<br>Settings:<br>1. Do the AIR1 settings.<br>2. AON and Offload: enable. For required configurations, see Other settings.<br>3. Display: Turn ON display AON.<br>4. Offload: Enable offload.<br>Measure:<br>Disconnect USB and wait for display time out and measure power for 60 sec. |
| Time check and 5 second timeout | TCK1 | Test application:<br>1. Install apk: `com.qualcomm.qti.weartech.watchface.test.`<br>2. Select the digital, black and white font watchface, which updates every minute in UI application list.<br>Settings:<br>1. Do the AIR1 settings.<br>2. AON and Offload: enable, TP: disable, Sensor: disable.<br>`adb shell settings put system screen_off_timeout 5000`<br>Measure:<br>Measure for 30 sec. While measuring power, press power button to wake up display, display turns OFF after 5s. |

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| Test case | Case ID | Test conditions |
|---|---|---|
| Offloaded Time Check and 5-second timeout | TCK1-2 | Test application:<br>1. Install apk: `com.qualcomm.qti.weartech.watchface.test.`<br>2. Select the digital, black and white font watchface, which updates every minute in UI application list.<br>Settings:<br>1. Do the AIR1 settings.<br>2. AON and Offload: enable, TP: disable<br>3. Offload: enable.<br>4. Tilt: enable.<br>5. Sensor: enable.<br>`adb root && adb remount`<br>`adb reboot`<br>`adb root && adb remount`<br>`adb push <<32.ASPEN\WearTechWatchFaceTest\WearTechWatchFaceTest.apk>> /system_ext/app/WearTechWatchFaceTest`<br>`adb reboot`<br>`adb shell pm enable com.qualcomm.qti.weartech.watchface.test`<br>Measure:<br>Measure for 30 sec. While measuring power, sensor (tilt) to wake up display, display turns OFF after 5s. |
| 3-layer UI | UI3L | Settings:<br>1. Do the AIR1 settings.<br>2. Display: select 10min for screen timeout.<br>Test application:<br>1. `adb install <<\Gears_v20180602_apkpure.com.apk>>`<br>2. Download `Gears_v20180602_apkpure.com.apk` from https://apkpure.com/gears/com.jeffboody.GearsES2eclair and install using the command: `adb install <<\Gears_v20180602_apkpure.com.apk>>`<br>3. Run Gear standby.<br>Measure:<br>Disconnect the USB and measure power for 60s. |

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| Test case | Case ID | Test conditions |
|---|---|---|
| Manhattan 3.0 1080p off screen 9.6fps | MAN1080p | Settings:<br>1. Do the AIR1 settings.<br>2. Video tools: play Manhattan 3.0 1080p off screen 9.6fps.<br>`adb root`<br>`adb remount`<br>`adb install --bypass-low-target-sdkblock <<gfxbench_vulkan-5.0.0-rfix+corporate.arm64-v8a.armeabiv7a.x86.x86_64.apk path>>`<br>`adb reboot`<br>3. Open GFXB application.<br>4. Accept the permission and enter the following command:<br>`adb shell am start net.kishonti.gfxbench.vulkan.v50000.cor porate/net.kishonti.app.MainActivity && adb shell am broadcast -a net.kishonti.testfw.ACTION_RUN_TESTS -n net.kishonti.gfxbench.vulkan.v50000.cor porate/net.kishonti.benchui.corporat e.C ommandLineSession -e test_ids vulkan_5_normal_off --ei -single_frame 20868 --ei -fsaa 0 --ei -frame_step_time -1 --ei -max_rendered_frames -1 --ef -fps_limit 9.6 --ei -offscreen_width -1 --ei -offscreen_height -1`<br>5. Disconnect the USB and collect waveforms.<br>Note: Measure at 9.6 fps result. |

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| Test case | Case ID | Test conditions |
|---|---|---|
| Manhattan 3.0 1080p off screen 60fps | MAN1080p | Settings:<br>1. Do the AIR1 settings.<br>2. Video tools: play Manhattan 3.0 1080p off screen 60 fps.<br>`adb root`<br>`adb remount`<br>`adb install --bypass-low-target-sdkblock <<gfxbench_vulkan-5.0.0-rfix+corporate.arm64-v8a.armeabiv7a.x86.x86_64.apk path>>`<br>`adb reboot`<br>3. Open GFXB application.<br>4. Accept the permission and enter the following command:<br>`adb shell am start net.kishonti.gfxbench.vulkan.v50000.cor porate/net.kishonti.app.MainActivity && adb shell am broadcast -a net.kishonti.testfw.ACTION_RUN_TESTS -n net.kishonti.gfxbench.vulkan.v50000.cor porate/net.kishonti.benchui.corporat e.C ommandLineSession -e test_ids vulkan_5_normal_off --ei -single_frame 20868 --ei -fsaa 0 --ei -frame_step_time -1 --ei -max_rendered_frames -1 --ef -fps_limit 60 --ei -offscreen_width -1 --ei -offscreen_height -1`<br>5. Disconnect the USB and collect waveforms.<br>Note: Measure at 60 fps result. |
| 1080p Video decode H.264 30fps , displayed on 480x480 screen | QTC77 | Settings:<br>1. Do the AIR1 settings.<br>2. Use Album to play 1080p Video decode H.264 30fps.<br>3. Push 10800p videoclip to `/sdcard`.<br>`adb push <<path of 1080P_videoclip>>\Qtc88_91sec.mp4 /sdcard/`<br>`adb reboot`<br>4. Open Album to play video.<br>Measure:<br>Measure power 30s during the test and record the GPU render FPS output from test result. |

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| Test case | Case ID | Test conditions |
|---|---|---|
| Camera Preview 4640 * 3488 (12MP Sensor), 30fps , no EIS | QMC38A-1 | Settings:<br>1. Do the AIR1 settings.<br>2. Display: select 10min for screen timeout.<br>3. Camera settings:<br>a. Launch the snapdragon camera apk in the photo Preview mode.<br>b. Open Settings menu.<br>c. Disable "Face Detection" and other options use the following default settings:<br>i. Confirm camera parameters in "developer options".<br>ii. Continuously click 7 times on "version info" to enter the develop mode.<br>iii. Confirm "Picture size" uses 4640 * 3488 (12MP Sensor), 30fps , no EIS.<br>iv. Confirm "ZSL" uses default setting "HAL-ZSL".<br>Measure:<br>Disconnect USB and measure power for 60 seconds. |
| Camera encode H264_ AAC 12MP sensor,VGA enc,30fps no EIS | VGA30 | Settings:<br>1. Do the AIR1 settings.<br>2. Display: select 10min for screen timeout.<br>3. Camera:<br>a. Launch the snapdragon camera apk in the Video mode.<br>b. Configure the camera video parameters.<br>c. 12MP sensor,VGA encode, 30fps no EIS<br>d. Other options use default settings<br>e. Start recording and turn OFF the microphone.<br>Measure:<br>Disconnect USB and measure power for 60 seconds. |
| Camera encode H264 AAC 12MP sensor,1080p enc,30fps , no EIS | HD30ENOEIS | Settings:<br>1. Do the AIR1 settings.<br>2. Display: select 10min for screen timeout.<br>3. Camera:<br>a. Launch the snapdragon camera apk in the Video mode.<br>b. Disable "Face Detection". Other options use the following default settings:<br>c. Confirm the camera parameters in "developer options".<br>d. Continuously click 7 times on "version info" to enter the develop mode.<br>e. 12MP sensor,1080p encode,30fps , no EIS.<br>Measure:<br>Disconnect USB and measure power for 60 seconds. |

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| Test case | Case ID | Test conditions |
|---|---|---|
| LTE FDD 20MHz CDRX Standby 320ms long CDRX cycle 1Rx | CDRXRS5 | ▪ NR SA Sub6 FDD CDRx, Long cycle=320ms, On duration=10ms, SSB -PO offset=0ms.<br>▪ UE should be in 1Rx, No neighbors. Backlight and display off.<br>Measure:<br>1. After DUT camped to LTE network, disconnect USB.<br>2. Wait for screen timeout<br>3. Take power measure for 60s. |
| LTE FDD 20MHz Standby 1.28s | LS11 | ▪ NR SA Sub6 FDD Standby mode, DRX=1.28sec, SSB -PO offset=10ms RxAGC at phone ~ -50 dBm.<br>▪ UE should be in 1Rx, No neighbors. Backlight and display off.<br>▪ Network type : LTE<br>Measure:<br>▪ Take power measure for 60s. |
| LTE FDD 20MHz 1Rx PDCCH Only, Band4 | LTE17E-1 | ▪ FDD LTE<br>▪ Bandwidth: 20MHz<br>▪ Mac Padding OFF<br>▪ UE Settings:<br>▪ Disable Airplane mode<br>Measure:<br>▪ Take power measure for 60s. |
| LTE FDD CAT1 20MHz 1Rx Data call, Tx+0dBm, RSRP-105dBm<br>(RB mode) (DL: 10Mbps, UL: 5Mbps) | LTE5RB18 | ▪ FDD LTE<br>▪ Bandwidth: 20MHz<br>▪ RSRP: -105dBm<br>▪ Tx 0dBm<br>▪ DL: 10Mbps, UL: 5Mbps, RB mode (Mac Padding ON)<br>▪ Set APN<br>Measure:<br>▪ Take power measure for 60s |
| LTE FDD CAT4 20MHz 1Rx Data call, Tx+0dBm, RSRP-105dBm<br>(RB mode) (DL: 75Mbps, UL: 50Mbps) | LTE7RB18 | ▪ FDD LTE<br>▪ Bandwidth: 20MHz<br>▪ RSRP: -105dBm<br>▪ Tx 0dBm<br>▪ DL: 10Mbps, UL: 5Mbps, RB mode (Mac Padding ON)<br>▪ Set APN<br>Measure:<br>▪ Take power measure for 60s. |

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| Test case | Case ID | Test conditions |
|---|---|---|
| Background SMS receive (no notification) over LTE FDD 20 MHz measured over 60 sec | MRL1 | ▪ LTE FDD Standby mode, DRX=320 msec, 100 DL, 24 UL, B1,<br>▪ Inactivity timer – 10 sec, Rx power level -50 dBm at phone antenna connector.<br>▪ Measurement duration 60 sec.<br>▪ Sintra = 0 and snonintra = 0, no neighbor cells.<br>▪ Backlight and display off.<br>Procedure:<br>1. Measure for 60 sec.<br>2. During this period , send SMS 2 times from Callbox, at 15s and 45s after starting data acquisition respectively,<br>Note:<br>1. Before saving the test data, disable the SMS reminder and sounds. This method is send one message first. When SMS is received, slide right on the message on notification interface, then disable sound and reminder.<br>2. SMS send at an interval of 30s on Callbox manually. |
| Video call over LTE FDD CAT4 20MHZ 16MP Camera 30fps Tx + 0 dBm w/ EIS, 480p Enc & Dec RSRP -90dBm | VIDCALL30 | ▪ Set video encoding to 480p.<br>▪ Use 12MP front camera.<br>▪ Internet Environment Setup.<br>▪ See the guide on configuring the WhatsApp network environment.<br>▪ Network Connection Check.<br>▪ Connect DUT (Device Under Test) to the network.<br>▪ Make a Video Call.<br>▪ Install WeChat using ADB:<br>`adb install ""<<path of WeChat\Weixin 8.0.60.apk>>""`<br>▪ Use the other phone as the REF device.<br>▪ Keep mobile data ON.<br>▪ Disable Wi-Fi.<br>▪ Douyin (TikTok) Video Call Test (Optional).<br>▪ Install Douyin using ADB:<br>`adb install ""Path of WeChat\douyin.apk""`<br>▪ Use SMS verification to log in.<br>▪ After login, initiate a video call with your own account.<br>▪ Start Acquisition.<br>▪ Disconnect the USB cable.<br>▪ Start data acquisition for 60 seconds. |

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| Test case | Case ID | Test conditions |
|---|---|---|
| Audio streaming over LTE FDD 10MHZ CAT1 1Rx Tx +10dBm, RSRP -70dBm to BTHS BT Tx +<br>4 dBm(Measured over 15-20 min) | AS02AWLBT | ▪ DL: MCS:6, RB 50<br>▪ UL: MCS:3, RB 24<br>▪ RSRP -70dBm<br>▪ P-MAX： 10<br>UE side settings:<br>▪ connect RF cable, enable Bluetooth and connect to Bluetooth headset.<br>`app install: adb install <<Path ..\01.AS02AWLBT_5100.APK\app-debug.apk>>`<br>▪ Launch music play app and download music file from http server to trigger loop play:<br>`adb shell am instrument -e class com.qualcomm.qti.aptmm_mediaplayer.AudioTest\#testLoopPlayStreamingAudio -e looptime 100 -e mediafile_param http://192.168.2.1/media/Richard.aac com.qualcomm.qti.aptmm_mediaplayer/.MediaPlayerTestRunner`<br>▪ Disconnect USB cable and display off, wait 30s and then start power measurement and waveform capture.<br>Measure:<br>▪ Power measurement for this use-case needs to be done for 15 minutes to get the data |
| VoLTE FDD 20MHz Voice call - CAT1 1Rx Tx +0dBm, RSRP -90dBm, (40%Talk,40%listen, 20% silence),<br>CDRX On 40 milliseconds | VoLTEW3 | ▪ Airplane mode disable, disable Wi-Fi/NFC/ location/date and time, disable data, enable VoLTE<br>▪ UI setting > cellular > select network type LTE Only; set APN to Keysight<br>▪ DUT connect with Bluetooth.<br>▪ cell on and connect RF cable to UE PRX ANT port, attach to callbox LTE network(HD)<br>▪ For VoLTE Audio Test File, see *VoLTE Audio Test File For Power Measurement* (MH80-VR010-21).<br>▪ Make phone call (40%Talk,40%listen, 20% silence)<br>▪ Take Power measurement for 60s |
| Sub6 SA FDD 20MHz VoNR, 1RxTx+0dBm, RSRP -90 dBm<br>(40% Talk, 40% Listen,20% Silence),CDRX ON 40ms | VoNR | ▪ Sub6 SA FDD 20MHz Voice overNR, Tx= +18dBm, RSRP -90 dBm<br>▪ UE should be in 1Rx, No neighbors, Backlight and display off.<br>Procedure:<br>▪ Talk state – Audio clip is played at the UE DUTwhile the remote device is muted. For more details, see *VoLTE Audio Test File for Power Measurement* (MH80-VR010-21)<br>▪ ▪ Listen state – DUT is muted and the audio clip is played at the remote device. For more details, see *VoLTE Audio Test File for Power Measurement* (MH80-VR010-21).<br>▪ Silent state – Both the DUT and the remote device should be muted.<br>▪ Measure for 60 sec.<br>Note: For VoNR test case, see *VoLTE Audio Test File for Power Measurement* (MH80-VR010-21). |

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| Test case | Case ID | Test conditions |
|---|---|---|
| Video Call over 5G sub6 1Rx 12MP Camera 30fps Tx +0dBm, RSRP -90dBm, 30fps 480p enc/dec | VIDCALL5G_30 | ▪ RSRP: -90 dBm<br>▪ Tx Power: +18 dBm<br>▪ Disable Airplane Mode<br>▪ Disable the following features: Bluetooth/Wi-Fi/NFC/ Location<br>▪ Automatic Date and Time set to: NR Only<br>▪ Turn Mobile Data ON<br>▪ Set APN to Keysight<br>▪ Configure video settings:<br>▪ 480p Encoding<br>▪ 12MP Front Camera<br>▪ Internet Environment Setup<br>▪ See the WhatsApp network environment configuration guide.<br>▪ Connect DUT to the network<br>▪ WeChat Video Call Test<br>▪ Install WeChat via ADB:<br>▪ Keep Mobile Data ON<br>▪ Disable Wi-Fi<br>▪ Douyin (TikTok) Video Call Test (Optional)<br>▪ Install Douyin via ADB:<br>▪ After login, initiate a video call with your own account<br>Measurement<br>▪ Disconnect the USB cable<br>▪ Start power acquisition for 60 seconds |
| Audio streaming over 5G, 1RxTx+10dBm, RSRP -70dBm, BT HS, BTTx +10dBm max | AS02AW5GBT | ▪ RSRP -70dBm<br>▪ P-MAX： 10<br>▪ Connect RF cable, enable Bluetooth and connect to bluetooth headset.<br>▪ app install:<br>`<<Path ...01.AS02AWLBT_5100.APK\app-debug.apk>>`<br>▪ Launch music play application and download music file from http server to trigger loop play:<br>`adb shell am instrument -e class com.qualcomm.qti.aptmm_mediaplayer.AudioTest\#testLoopPlayStreamingAudio -e looptime 100 -e mediafile_param http://192.168.2.1/media/Richard.aac com.qualcomm.qti.aptmm_mediaplayer/.MediaPlayerTestRunner`<br>▪ disconnect USB cable and display off, wait 30s and then start power measurement and waveform capture.<br>Measurement:<br>Power measurement for this use-case needs to be done for 15 minutes to get the data. |

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| Test case | Case ID | Test conditions |
|---|---|---|
| sub6 FDD 20MHz standby 1.28s 10ms SSB-PO offset 1Rx | TBD1 | ▪ Set SSB-PO Offset to 10ms<br>▪ Paging Frame Offset: 1<br>▪ Set Default Paging Cycle to 1.28s (128)<br>Note: After network connection, manually release the RRC connection.<br>▪ Disable Airplane Mode<br>▪ Set to: NR (5G)<br>Measurement Procedure<br>▪ Perform power measurement for 60 seconds |
| sub6 FDD 20MHz CDRX 320ms DRX ,10ms on duration, 0ms SSB offset 1Rx | TBD2 | ▪ "0ms SSB offset：(SSB→ Periodicity:20ms DRX→ Long Cycle:ms320<br>▪ Disable Air-plane mode<br>▪ Settings > Network&internet > SIMs > advanced > Preferred Network type : NR<br>Measure:<br>▪ Take power measure for 60s" |
| Sub6 SA FDD PDCCH only 20MHz 1Rx | TBD3 | ▪ Disable Air-plane mode<br>▪ Network type : NR<br>Measure:<br>▪ Take power measure for 60s |
| Sub6 FDD Data 20MHZ 1Rx 75/48Mbps , Tx +0dBm, RSRP -105dBm (RB Test) | TBD4 | ▪ DL MCS/RB to 27/100, UL MCS/RB to /100<br>▪ Disable Air-plane mode<br>▪ Network type : NR<br>Measure:<br>▪ Take power measure for 60s" |
| GNSS standalone 1 Hz tracking [Non-DPO] | GNSS1-2 | Precondition:<br>▪ Do the AIR1 settings.<br>▪ Bluetooth, Wi-Fi, NFC, Display, Sound – OFF,<br>▪ Gestures - OFF<br>▪ NV70326 - 0x8851<br>▪ NV05596 – 0, NV74255-0<br>▪ Xtra – Available<br>Procedure:<br>1. Connect the GNSS live RoofTop antenna feed.<br>2. Apply NV settings according to the use case (a reboot is required after setting NV)<br>3. Start the GNSS engine from QXDM, go to view folder on right side of QXDM, write CGPS GPS Window pops up and click run GPS, use case starts.<br>4. Disconnect the QXDM Professional™ Tool.<br>5. Disconnect the USB from power measurement software.<br>6. Start power measurement.<br>7. Measure current for 60 sec.<br>Device configuration:<br>1. By default, `izat.conf` is set to disable. To ensure it is disabled, check the config using command: `adb pull /vendor/etc/izat.conf`.<br>2. If it is not disabled, set it disable using `SAP=Disabled`.<br>3. Push in the same location `/vendor/etc/` and reboot the device. |

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| Test case | Case ID | Test conditions |
|---|---|---|
| GNSS 1Hz L1L5 Trk (NON-DPO) | GNSS4-2 | Precondition:<br>1. Do the AIR1 settings.<br>2. Bluetooth, Wi-Fi, NFC, Display, Sound – OFF,<br>3. Gestures - OFF"<br>4. NV70326 - 0x8851<br>5. NV05596 – 0, NV74255-39<br>6. Xtra – Available<br>Procedure:<br>7. Connect the GNSS live RoofTop antenna feed.<br>8. Apply NV settings according to the use case (a reboot is required after setting NV)<br>9. Start the GNSS engine from QXDM, go to view folder on right side of QXDM, write CGPS GPS Window pops up and click run GPS, use case starts.<br>10. Disconnect the QXDM Professional™ Tool.<br>11. Disconnect the USB from power measurement software.<br>12. Start power measurement.<br>13. Measure current for 60 sec.<br>Device configuration:<br>1. By default, `izat.conf` is set to disable. To ensure it is disabled, check the config using command: `adb pull /vendor/etc/izat.conf`.<br>2. If it is not disabled, set it disable using `SAP=Disabled`.<br>3. Push in the same location `/vendor/etc/` and reboot the device. |

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| Test case | Case ID | Test conditions |
|---|---|---|
| GNSSL1ACQL5TRK | GNSS3-1d | Precondition:<br>1. Do the AIR1 settings.<br>2. Bluetooth, Wi-Fi, NFC, Display, Sound – OFF<br>3. Gestures - OFF<br>4. NV70326 - 0x8851<br>5. NV05596 – 0, NV74255-39<br>6. Xtra – Available<br>7. Terminate the GNSS drop using a 50-ohm terminator (this step is applicable only for the GNSS L1 ACQL5TRK use case)<br>Procedure:<br>8. Connect the GNSS live RoofTop antenna feed.<br>9. Apply NV settings according to the use case (a reboot is required after setting NV)<br>10. Start the GNSS engine from QXDM, go to view folder on right side of QXDM, write CGPS GPS Window pops up and click run GPS, use case starts.<br>11. Disconnect the QXDM Professional™ Tool.<br>12. Disconnect the USB from power measurement software.<br>13. Start power measurement.<br>14. Measure current for 60 sec.<br>Device configuration:<br>1. By default, `izat.conf` is set to disable. To ensure it is disabled, check the config using command: `adb pull /vendor/etc/izat.conf`.<br>2. If it is not disabled, set it disable using `SAP=Disabled`.<br>3. Push in the same location `/vendor/etc/` and reboot the device. |

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## Other settings

## Disable display AON:

```
adb root
adb shell settings put secure doze_enabled 0
adb shell settings put secure doze_always_on 0
adb  shell ""settings get secure doze_always_on
```

## Enable display AON:

```
adb shell settings put secure doze_enabled 1
```

## Disable offload:

```
adb root
adb  shell setprop persist.sys.offload.debug.disable true
adb  shell ""getprop persist.sys.offload.debug.disable""
adb reboot
```

### Enable offload:

```
adb  shell setprop persist.sys.offload.debug.disable false
adb reboot
```

### Disable touch panel:

```
adb root
adb shell
echo 1 > sys/bus/i2c/devices/9-0039/raydium_i2c_touch_lock
```

### Enable CPU low power mode:

```
echo 0 > /sys/devices/system/cpu/qcom_lpm/parameters/sleep_disabled
```

### Disable sensor:

```
adb root
adb remount
adb reboot
adb wait-for-device root
adb remount
adb shell rm -rf /vendor/etc/sensors/hub2/config
adb shell rm -rf
/mnt/vendor/persist/sensors/hub2/sensors/registry/registry/*
adb shell dumpsys sensorservice restrict .cta
adb reboot
```

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# 4 Chipset current consumption

### Commercial software release target values

Commercial software release target values are predictions of the current consumption for the first commercial quality software release for an OS. These targets will not be updated after the commercial software has been released.

## 4.1 Power dashboard CS goals

Commercial software release target values for the operational modes are defined in Table 4-2. These values reflect optimized hardware and software configurations and are normalized to 3.7 V supply voltage. A little deviation on customers devices is expected from mentioned goals; therefore, RMA decisions must not be based on the results of any power measurements described in this document. Legitimate candidates for power consumption RMAs require devices that violate the maximum power specifications provided in *SW6100-0/SW6100P-0 Data Sheet* (80-83950-1).

**Table 4-1 Power dashboard CS goals – Top level**

| Area | Test Case | TCID | Tj (°C) | CS goals 4GB LPDDR5x Goal (mW) |
|---|---|---|---|---|
| Sleep | MSM Airplane mode sleep W/ display, Sensors, KWD, audio, tilt, touch, ALL OFF , DDR in SR | AIR1 | 25 | 7.18 |
| Display | Display ON (Static display w/ Smart panel) | LCD04A-C | 25 | AIR1 + 0.33 |
| | Static Display with Clock Update Once a minute | LCDCU1 | 25 | LCD04A-C+0.03 |
| | Static Display with Clock Update Once a second | LCDCU60 | 25 | LCD04A-C+0.98 |
| UI | Time Check and 5 sec timeout | TCK1 | 25 | AIR1 + 15.4 |
| | Offload Time Check and 5 sec timeout | TCK1-2 | 25 | AIR1 + 3.74 |
| | 3layer UI/gears,display update 60fps | UI3L | 25 | 92 |
| GFX | Manhattan 3.0 1080p off screen 9.6fps | MAN1080p | 25 | 310 |

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| Area | Test Case | TCID | Tj (°C) | CS goals 4GB LPDDR5x Goal (mW) |
|---|---|---|---|---|
| | Manhattan 3.0 1080p off screen 60fps | MAN1080p | 25 | 1500 |
| Video | 1080p Video decode H.264 30fps , displayed on 480x480 screen (348x448 on sw5100) | QTC77 | 25 | 191 |
| Camera | Camera Preview 4640 * 3488 (12MP Sensor), 30fps , no EIS | QMC38A-1 | 25 | 310 |
| | Camera encode H264_ AAC 12MP sensor,480p enc,30fps no EIS | VGA30 | 25 | 330 |
| | Camera encode H264 AAC 12MP sensor,1080p enc,30fps , no EIS | HD30ENOEIS | 25 | 419 |
| 4G Standby | LTE FDD 20MHz CDRX Standby 320ms long CDRX cycle 1Rx | CDRXRS5 | 25 | AIR1 + 8.9 |
| | LTE FDD 20MHz + Standby 1.28s | LS11 | 25 | AIR1 + 1.8 |
| 4G Data | LTE FDD 20MHz 1Rx PDCCH Only, Band4 | LTE17E-1 | 25 | 128 |
| | LTE FDD CAT1 20MHz 1Rx Data call, Tx+**0dBm**, RSRP-105dBm (RB mode) (DL: 10Mbps, UL: 5Mbps) | LTE5RB18 | 25 | 340 |
| | LTE FDD CAT4 20MHz 1Rx Data call, Tx+**0dBm**, RSRP-105dBm (RB mode) (DL: 75Mbps, UL: 50Mbps) | LTE7RB18 | 25 | 350 |
| | Background Msg Receive over LTE FDD 20MHz with display off | MRL1 | 25 | AIR1 + 24 |
| 4G Video call | Video call over LTE FDD CAT4 20MHZ 16MP Camera 30fps Tx + **0 dBm** w/ EIS, 480p Enc & Dec RSRP -90dBm | VIDCALL30 | 25 | 588 |
| 4G Audio | Audio streaming over LTE FDD 10MHZ CAT1 1Rx Tx +10dBm, RSRP -70dBm to BTHS BT Tx + **4 dBm**(Measured over 15-20 min) | AS02AWLBT | 25 | 77 |
| 4G Voice | VoLTE FDD 20MHz Voice call - CAT1 1Rx Tx **+0dBm**, RSRP -90dBm, (40%Talk,40%listen, 20% silence), CDRX On 40 milliseconds | VoLTEW3 | 25 | 155 |
| 5G Voice | Sub6 SA FDD 20MHz VoNR, 1RxTx+**0dBm**, RSRP -90 dBm (40% Talk, 40% Listen,20% Silence),CDRX ON 40ms | VoNR | 25 | 150 |
| 5G Videocall | Video Call over 5G sub6 1Rx 12MP Camera 30fps Tx **+0dBm**, RSRP -90dBm, 30fps 480p enc/dec | VIDCALL5G_30 | 25 | 578 |
| 5G Audio | Audio streaming over 5G, 1RxTx+10dBm, RSRP -70dBm, BT HS, BTTx +10dBm max | AS02AW5GBT | 25 | 77 |
| 5G Standby | sub6 FDD 20MHz standby 1.28s 10ms SSB-PO offset 1Rx | TBD1 | 25 | AIR1 + 2.0 |
| | sub6 FDD 20MHz CDRX 320ms DRX ,10ms on duration, 0ms SSB offset 1Rx | TBD2 | 25 | AIR1 + 7.7 |

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| Area | Test Case | TCID | Tj (°C) | CS goals 4GB LPDDR5x Goal (mW) |
|---|---|---|---|---|
| 5G Data | Sub6 SA FDD PDCCH only 20MHz 1Rx | TBD3 | 25 | 126 |
| | Sub6 FDD Data 20MHZ 1Rx 75/48Mbps , Tx **+0dBm**, RSRP -105dBm (RB Test) | TBD4 | 25 | 340 |

## 4.2 Power dashboard ES1 measurements

**Table 4-2 Power dashboard ES1 measurements**

| Test case | TCID | SW6100 + 4GB LPDDR5x Adjusted Measure in ES1 software version (mW) |
|---|---|---|
| MSM Airplane mode sleep with display, sensors, KWD , audio , tilt , touch, ALL OFF , DDR in SR | AIR1 | 18.37 |
| Time check and 5 second timeout | TCK1 | 57.60 |
| Manhattan 3.0 1080p off screen 9.6fps | MAN1080p | 426.11 |
| Camera Preview 4640 * 3488 (12MP Sensor), 30fps, no EIS | QMC38A-1 | 434.65 |
| LTE FDD 20MHz 1Rx PDCCH Only, Band4 | LTE17E-1 | 174.95 |

## 4.3 Breakdown measurement per regulator values

This section lists the rail level breakdown data for all use cases on power dashboard based on the build for the ES1 release.

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## Modem and multimedia use cases

To view the rail-level breakdown data for modem and multimedia use cases, select SW6100_LAW1.0.xlsx from the attachments panel or click the icon.

## Adjustment rails

To view the rail-level breakdown data for adjustment rails, select adjustment.xlsx from the Attachments panel, or click the icon.

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# A References

## A.1 Related documents

| Title | Number |
|---|---|
| **Qualcomm Technologies, Inc.** | |
| *SW6100-0/SW6100P-0 Data Sheet* | 80-83950-1 |
| *Configuration of Input Pins During Device Sleep* | 80-VN499-7 |
| *VOLTE Audio Test File for Power Measurement* | MH80-VR010-21 |

## A.2 Acronyms and terms

| Acronym or term | Definition |
|---|---|
| AP | Access point |
| CS | Commercial sample |
| DoU | Days of use |
| IS | Image stabilization |
| RCM | Resistive current measurement |
| WTP | Wearable test platform |
| ZSL | Zero shutter lag |

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